

The Patch Test: A Comprehensive Calibration Tool for Multibeam Echosounders

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Abstract— The patch test procedure is the first stage required in any multibeam echosounder survey. The patch test is designed to precisely determine the static configuration of the sonar head (roll, pitch, yaw) and the latency remaining between the reception of the GPS fix and its integration by the acquisition system. An imprecise knowledge of these four values introduces bias when computing the real 3D geographic location of each beam.

In the proposed patch test program, these four values are sequentially computed, according to tailored track acquisitions. An extended set of measured information is involved, such as the dynamic attitude, position and heading of the support-ship, the geometric offsets of each acquisition unit (MRU, GPS, MBES, compass) and some environmental features (sound velocity profile). The computation mainly relies on performing a matching between specific sea-bottom profiles extracted from each overlapping track. A first rough computation is performed to help the user quickly select the range and step of values to search. Then the user can refine its search as required with successive iterations, during which both the error measures and the error curve for all the previous tested values are displayed. The computation is thus fully automatic, searching the best value within the range. The impact of selecting such a value is immediately shown on the global display of the surveyed area, which provides three representations to enable the user to evaluate the accuracy of the current result: a color-coded real location of the beams, the iso-depth contours extracted from each track, and synthesized depth grids which perform direct comparisons between the track depths. The user can then check if the accuracy he or she requires has been fulfilled.

Keywords— Bathymetry Data, Calibration Procedure, Patch Test.

INTRODUCTION

THE RESON Patch Test Program is a calibration tool for RESON Multibeam Echo Sounders (MBES). The program is used to measure the installation offsets of a MBES sonar head in relation to its associated external

sensors: motion sensor (roll and pitch), gyro (yaw), DGPS (latency). It is essential that this calibration be accomplished prior to any data processing.

The Patch Test program compares the bathymetry data between sets of overlapping calibration lines. Four offsets are determined: the static configuration of the sonar head (roll, pitch and yaw) and the latency remaining between the reception of the GPS fix and its integration by the acquisition system. After a validity check of the lines, the program will automatically apply a profile selection, a rough estimation, and a trial-and-error procedure to develop the optimum estimate of each offset.

I. MBES CALIBRATION

This section provides information on setting up a patch test and an explanation on how to acquire the proper data for use with the RESON Patch Test Program [1].

A. Why and how ?

To accurately acquire the seafloor, the measurements made by a multibeam sonar must be relative to the true vertical as reported by the motion sensor, and to the heading reported by the gyro compass. When the sensors are installed on a vessel, it is not possible to get them in perfect alignment. An additional calibration procedure, commonly referred to as a *patch test*, must be performed to derive actual offset values, which can then be applied to the data to bring the systems into proper alignment.

The procedure involves collecting data over certain types of terrain and processing it by means of a set of patch test tools. Currently, two primary methods of processing this data are used: an interactive graphical approach and an automatic, iterative surface match [1]. Each of these techniques has strengths and weaknesses and the preferred approach is dependent on the types of terrain features available to the surveyor.

The choice of which type of patch test tool should be used is a function of available features in the survey area. Graphical techniques work best when there are small, well defined features available (e.g. a one meter diameter rock or a pipe-line). A surface match is always preferred for roll offset determination, which requires a flat seafloor. A surface match is also preferred when the features are limited to slopping seafloor rather than discrete objects. The inherent averaging of larger numbers of comparisons

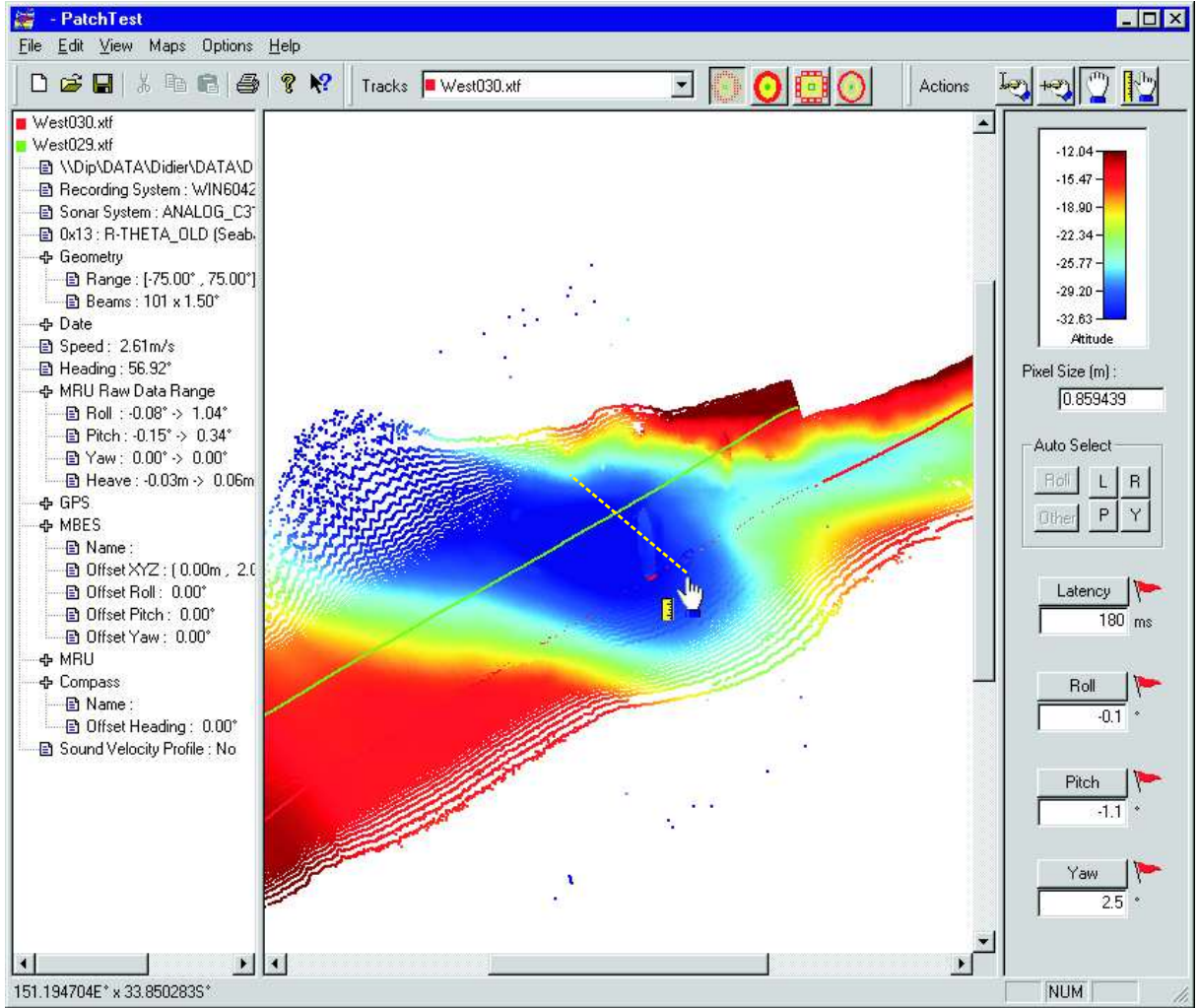


Fig. 1. RESON Patch Test main window

provides much more consistent results in these cases.

B. Static & dynamic roll, pitch and yaw

The orientation of the sonar head must be known in order to convert the measured slant ranges to depths and to determine the position of each of the determined depths.

Two kinds of attitude information are manipulated : static ones relative to the sonar head (they need to be determined) and dynamic ones measured on-line, by the MRU.

The orientation of the vessel supporting the sonar head, relative to the true vertical is measured by means of pitch and roll sensors. These sensors are typically an integral part of the MRU.

In order to use these data from the pitch and roll sensors, the alignment between the axes of the MRU and the sonar head must be known. This is usually achieved by ensuring that the housing of the MRU and the housing of the sonar head are aligned when the equipment is installed on the vessel or the ROV. Any misalignment will obviously cause the measured pitch and roll of the sonar

head to be in error.

Any error in the measured roll of the sonar head can cause substantial shifts in the conversion from slant range to depth (a roll error of 1° on a 50 meters slant range will cause a 0.6 meter difference in the resulting depth). Any error in the measured pitch of the sonar head will have a detrimental effect, primarily, on the accuracy of the positions that are determined for each slant range/depth. Nevertheless, the pitch error will have to be substantial before it has a recognizable impact on the determined depths (a pitch error of 1° will cause an along-track displacement in the position of 0.4 meter when the sonar head is 25 meters above the seabed).

Knowing the quality of the alignment between the sonar head and the axis of the MRU (and thus the vessel) is crucial. The patch test is designed to compute the residual not-known mis-alignment, i.e. the relative orientation of the sonar head, in the reference system $\mathcal{R}$ associated with the vessel. The orientation around the three axes of $\mathcal{R}$ is then determined by computing the pitch, roll and yaw of the sonar head. Roll is about the $\mathcal{R}_Y$ axis, which is the

along-ship axis; pitch is about the $\mathcal{R}_X$ axis, which is the across-ship axis; heave is along the $\mathcal{R}_Z$ axis.

II. THE PATCH TEST PROCEDURE

The physical offsets that must be determined are roll, pitch and yaw. In the case of a dual head sonar, each sonar head must be calibrated independently. If the data collection system is not synchronized to GPS time, it will also be necessary to determine the latency in the positioning system. Latency, as described here, is the time between the reported position being valid and the time the message is output and available to the system. Installation offsets are normally determined in a fixed sequence: latency (if required), roll, pitch and yaw.

If the MBES transducer is installed with an angular offset, such as in a typical dual-head installation, it is necessary either to manually apply the installation offset or perform roll calculations prior to latency calculations

A. Latency

As previously stated, the latency offset is the time interval between a position measurement being valid and the time that measurement is reported over the serial interface to the data collection system. In DGPS systems, this time normally ranges from 0.1 to 1.2 seconds, depending on the make, model, and mode of GPS receiver being used. Determination of the latency is required if the data collection system is not synchronized to GPS time and can not use the position time reported in the position message. If the data collection system is time synchronized, this step can be omitted.

RTK measurements can have much longer latencies and tend to have much more latency variation, measurement to measurement. As such, the use of a fixed latency value is normally not valid with RTK systems.

An incorrect latency results in a position error along track, that is a function of the latency value and the speed of the vessel.

DATA TO BE USED: To determine the latency, the surveyor will set up a survey line directly over a well defined feature, or perpendicular to a slope.

PROCEDURE: Two data sets will be collected, with the vessel traveling in the same direction, but at two significantly different speeds. One line should be collected at the minimum speed at which the vessel can maintain good steering. The other should be collected at the maximum survey speed. Only the data near nadir should be used for processing to minimize the effects of yaw offset.

The target resolution for the processing is 0.1 seconds. For high-speed survey vessels, a finer resolution may be required.

B. Roll offset

With a multibeam system, a roll offset results in depth measurement errors, which increase with both angle off

nadir and water depth.

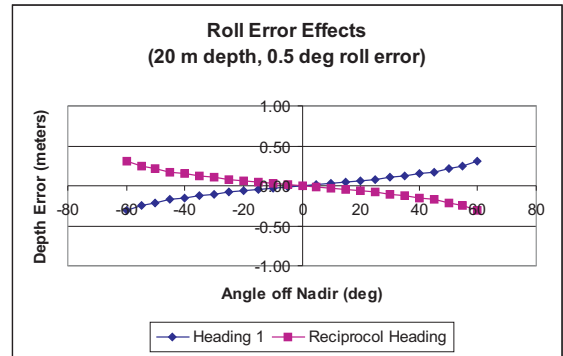


Fig. 2. Roll error effects

DATA TO BE USED: To determine the roll offset, the surveyor must collect two sets of data over a flat seafloor. On the same survey line, data collected on reciprocal headings will result in the same depth at nadir, but different depths away from nadir. When a cross section containing both data sets is viewed, any roll offset causes the data to appear as an “X”, as displayed in Figure 2.

PROCEDURE: The roll correction can be determined by applying varying roll offsets to the data sets and updating the display until the data from the two lines match. This can be done either by providing the operator with a graphical cross-section view and a set of controls to vary the roll offset, or using an automated process to compare the data from the two lines and iterating the roll until the two surfaces match.

The automatic approach is preferred and used in this version of the patch test program. The full swath coverage and overlap area along line, should be used for the surface comparisons. A resolution of 0.01° must be carried out through consecutive iterations.

C. Pitch offset

Pitch errors also result in position errors. Their amplitude is a function of both pitch offset error and water depth.

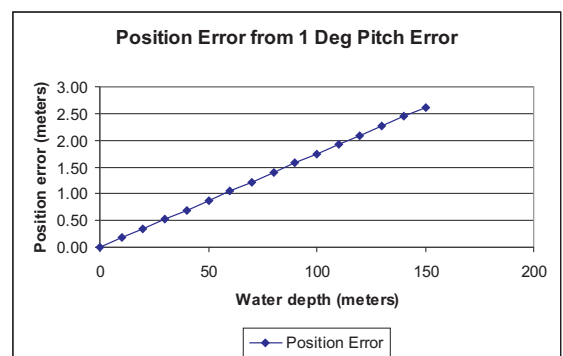


Fig. 3. Pitch error effects

DATA TO BE USED: Using the same line that was used

for the latency collection, the operator will collect two lines of data. These will be at the same speed, but in opposite directions on the survey line.

PROCEDURE: The processing and displays will be very much like those for latency. Again, only the data near the vessel track should be used in processing.

As can be seen in Figure 3, the errors induced by small pitch offsets in shallow water will very often be less than the accuracy of the positioning system. This may often prevent getting a reasonable and consistent answer for the pitch offset.

The target resolution for the pitch offset is 0.1° .

D. Yaw offset

Yaw offsets result from an angular difference in the horizontal plane, between the heading lubber line (along ship axis) of the compass and rotation angle about the Z-axis of the sonar projector. A yaw offset error will have no effect on the nadir beams, but will introduce position measurement errors on the outer beams. The magnitude of the position error will also increase with depth (Cf. Figure 4).

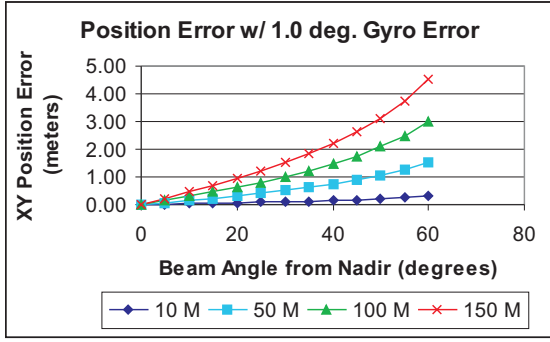


Fig. 4. Yaw error effects

DATA TO BE USED: To determine the yaw offset, the surveyor will set up two parallel survey lines over a well defined feature or perpendicular to a slope. The line spacing will allow the outer beams of one line to overlap the track for the other line (approximately 2 times the water depth). Tracks may be run either on reciprocal or same headings, with one track on each side of the object or feature.

PROCEDURE: The processing and displays will be very much like those for latency. If there is a yaw offset in the data, the position of a well-defined target will be different between the two data sets. If a slope is used, there will be shifts in the contour lines generated from the two data sets.

Any error in the measured heading of the sonar head will have a detrimental effect on the accuracy of the positions of the determined depths. For the center beams below the sonar head, this effect will be negligible, but will increase toward the outer beams. A heading error of

1° will cause an along-track shift in the position of 0.4 meter on the outer beams when the sonar head is vertically mounted and is 25 meters above the seabed.

III. THE RESON PATCH TEST PROGRAM

This section will describe some important aspects of the RESON Patch Test program, assuming some calibration datasets exist and comply with the constraints previously presented.

A. Computation

The goal of the patch test algorithm can be seen as a multi-criteria optimization problem. The four sought offsets constitute the solution $S = (S^1, S^2, S^3, S^4)$ to the problem $\langle \text{latency } (\kappa = 1), \text{roll } (\kappa = 2), \text{pitch } (\kappa = 3), \text{yaw } (\kappa = 4) \rangle$. Two main approaches can be investigated: one consists in searching simultaneously for the global 4-dimension solution [2], while other sequentially estimates the offsets in a specific order. Solving four mono-dimensional problems can be achieved because of the specific characteristics of the calibration tracks: the tracks acquired for computing the offset S^κ tend to minimize the effects of the other offsets on the data.

A.1 Generic algorithm

This basic algorithm estimates the matching error between selected surfaces for a given configuration S , in order to determine the best value for S^κ .

- Using the configuration S and the dynamic recorded attitude, compute the XYZ data D for the considered area. D_η contains the real 3D geographic location (XYZ) of the intersection of each beam of the track η with the sea-bottom.
- Working on these XYZ "surfaces", extract a series of I profiles P_η^κ , at the same location on each track η . Their characteristics (amount and direction in relation to the tracks) depend on the offset S^κ that is to be determined.
- For each set i of mated profiles, δ_i estimates the Z-mismatch error along the profiles belonging to this set.
- The last step consists in approximating the global surface error $E < S^\kappa >$ between the tracks within the considered area, by combining the δ_i errors.

$$E < S^\kappa > = \bigoplus_{i \in [1, I]} \delta_i \quad (1)$$

where $\bigoplus$ is the merging operator which assumes the profiles P^κ reliably sub-sample the original XYZ surfaces.

Iterations are made to loop over a range of values $[a, b]$. The best value found $\widetilde{S}^\kappa$ corresponds to the one which minimize $E < S^\kappa >$.

$$\widetilde{S}^\kappa = \min_{S^\kappa \in [a, b]} E < S^\kappa > \quad (2)$$

Table I indicates the features of the extracted profiles depending on the sought S^κ .

Offset	Features
Roll	P_{η}^1 : across tracks, along the selected area
Latency	P_{η}^2 : along tracks near their average nadir
Pitch	P_{η}^3 : along tracks near their average nadir
Yaw	P_{η}^4 : along tracks in the overlapping area of their outer swathes

TABLE I
PROFILES EXTRACTION CHARACTERISTICS

B. Results

This section will present results obtained with the calibration data acquired for the “Shallow Survey ’99” conference [3]. Each computation is preceded by the loading of appropriate files, an optional selection by the user of the areas that will be considered, and the launch of the computation itself by pushing the adequate button on the screen as shown in Figure 1.

B.1 Rough estimation

The first task of the program will be to determine a rough estimation of the sought offset. Several algorithms that directly compute the offsets, exist [4], taking into account various set of parameters. We choose to develop several equations that embed both the current dynamic attitude of the vessel, all the known static offsets and the already computed offsets. The goal is to guide further more precise computation, by initially refining the range of the offsets to be sought. Table II illustrates the rough estimation found for each offset and the proposed range to search, for determining the final precise value.

Offset	Rough estimation	Initial range	Final result
Roll	-0.3°	[-3.0°, 3.0°]	-0.1°
Latency	462ms	[-38ms, 962ms]	180ms
Pitch	-1.1°	[-4.0°, 2.0°]	-1.1°
Yaw	5.6°	[1.0°, 9.0°]	2.5°

TABLE II
EXAMPLE OF ROUGH ESTIMATIONS

B.2 Iteration scheme

- **LATENCY:** Figure 5 presents the interface that gathers all the results issued by each iteration. This window also allows the user to completely control the iteration scheme by defining the range and the step of the search. The left area of the window displays the state of the profiles involved in the current iteration, while the right panels store the already computed errors (in both curve and values “offset vs. error” shapes).

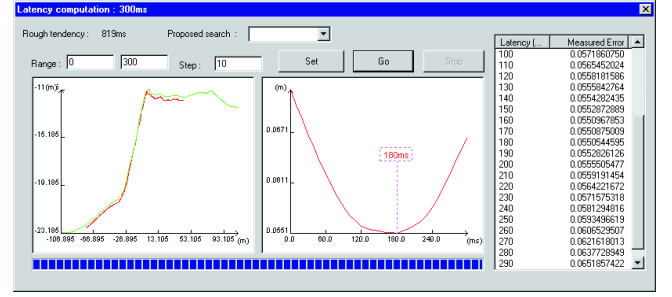
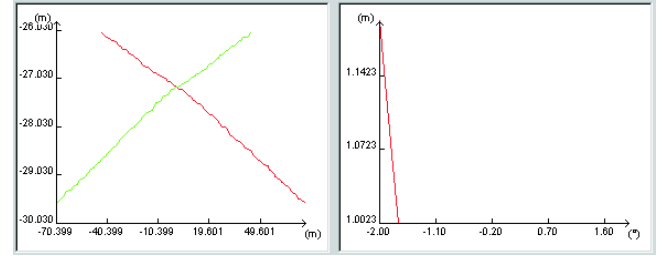
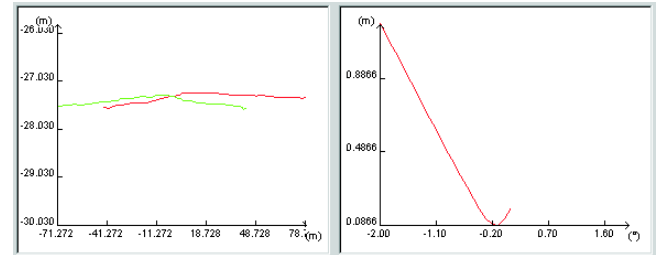


Fig. 5. Computing the latency

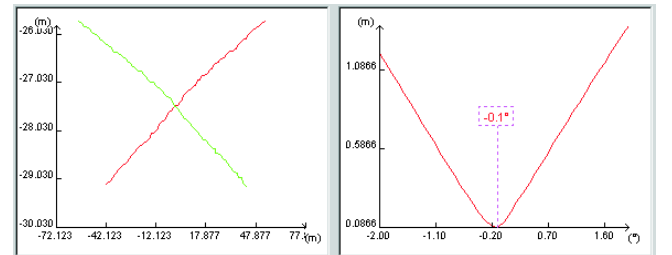
- **ROLL:** Figure 6 indicates the typical evolution of both the profiles and error curve when computing the roll offset. The range lies between -2.0° and 2.0°, with a 0.1° step. When the current offset used for the computation is far from the effective value, a classical X-shaped sea-bottom profiles is observed.



(a) Current roll offset: -1.7° (start)



(b) Current roll offset: 0.1° (just after the minimum)

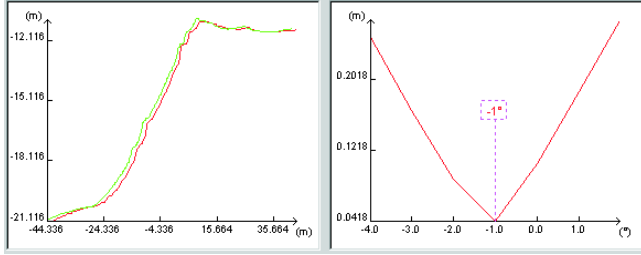


(c) Current roll offset: 2.0° (end)

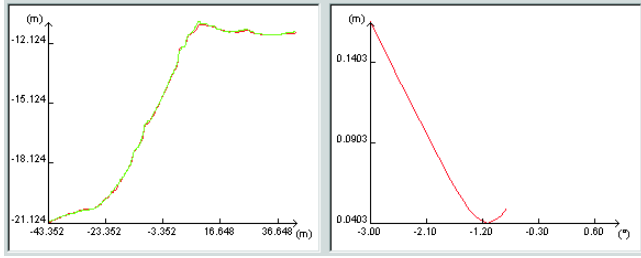
Fig. 6. Computing the roll offset

- **PITCH:** Figure 7 illustrates different resolution search of the pitch offset. The observed evolution of the profiles during a pitch computation is mainly done through

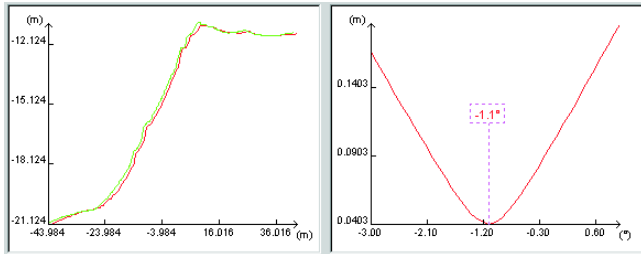
a translation.



(a) Rough search: $(-4.0^\circ, 2.0^\circ)$ (step: 1.0°)



(b) Fine search: $(-3.0^\circ, 1.0^\circ)$ (step: 0.1°)



(c) Fine search (end)

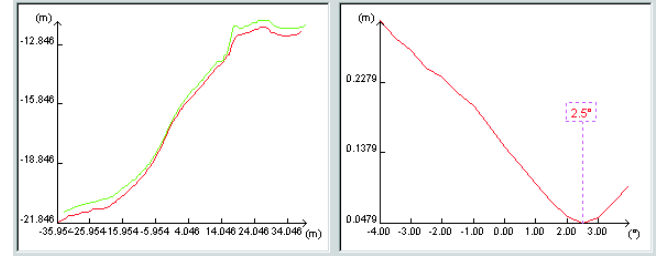
Fig. 7. Computing the pitch offset: refining the search

- **YAW:** Compared to the pitch step, the same kind of evolution for the profiles is observed, except the translation is replaced by a combination of translation and torsion, as the extracted profiles are located in the overlapping outer swath of the tracks.

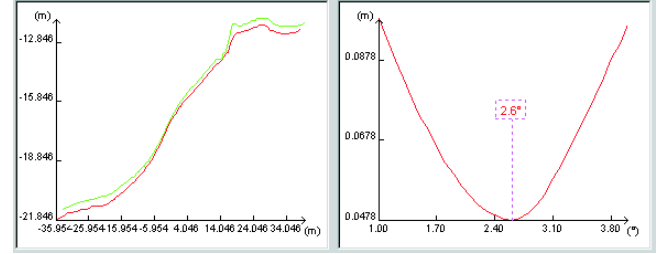
C. Results

Table II gathers the final offsets determined with the calibration dataset used by the “Shallow Survey ’99” conference [3]. The following results are obtained using the previously computed offsets on a subsequent survey. They will emphasize the impact of calibrating correctly the MBES system.

Two tracks, part of the Western survey, have been selected because of their coverage of a wide, two-meters height wreck. Figure 9 presents the XYZ coming from these two tracks, in two situations: without and with applying the offsets resulting from the patch test. The first case means all the offsets are set to 0. The contour of this wreck is outlined on both the tracks, and the plain lines indicates the paths of the sensor, for the two tracks



(a) Rough search $(-4.0^\circ, 4.0^\circ)$ (step: 1.0°)



(b) Fine search $(1.0^\circ, 4.0^\circ)$ (step: 0.1°)

Fig. 8. Computing the yaw

involved.

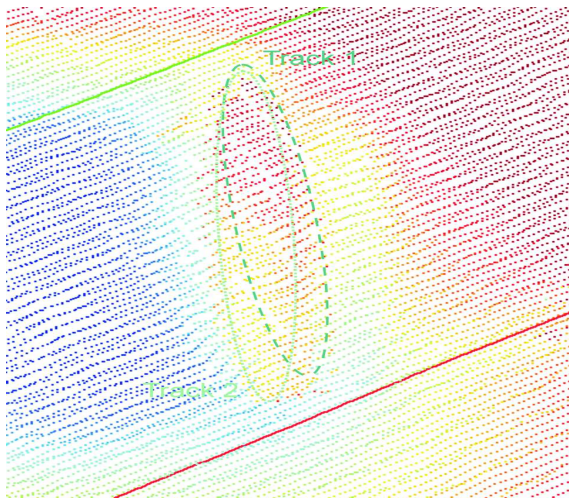
D. Interface

Besides the fully automatic computation of the offsets, the patch test interface goal is two-fold: first to allow a finer selection and tuning of the data involved in the computation, and second, to provide the user with tools permitting him to perform manual offsets simulations, or measures & verifications.

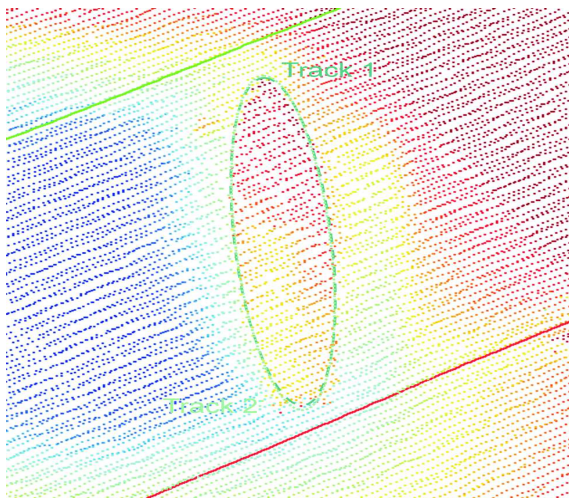
In the latter family, an obvious help is to be able at any time, to manually set roll, pitch, yaw and latency offsets and relaunch the XYZ data generation, through the right panel of the main window. Associated with a profile extraction tool (Cf. Figure 1) which operates directly on the display which gathers the whole XYZ dataset coming from each track, it allows to perform some checks and experiments (even a completely manual patch test). Nevertheless, the profile extraction is actually used to measure distance, slope and orientation, between points of the profiles, in order to have a precise knowledge of the relative location of each track.

Several other features are available when using the Patch Test program, either directly accessible in the main window or through various menus and dialog boxes. The following list will gather the fonctionnalités, the operator can expect to find and use, when performing bathymetry calibration with the Patch Test program.

- Adjustable color-coded depths of the XYZ data (selection of the colormap, and tuning of the range to focus on interesting features)
- Configurable warnings when the tracks involved in an offset computation do not comply with the acquisition



(a) Raw XYZ data

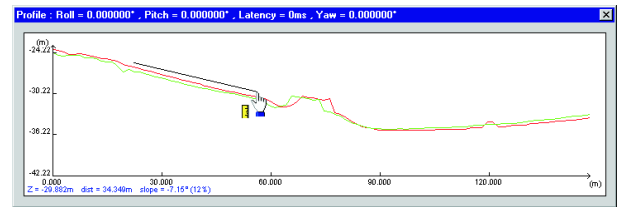


(b) After calibration

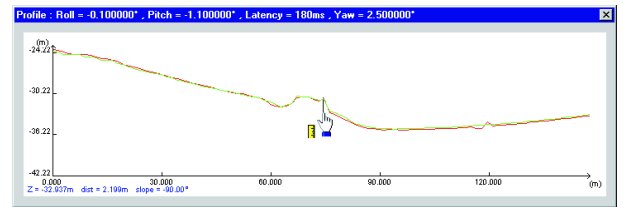
Fig. 9. Impact of the calibration procedure on the XYZ data

requirements corresponding to this offset.

- Extra synthesized representation for the XYZ raw data: iso-depth contours, grids merging the XYZ information in cells according to user selected operators (average, standard deviation, median, min-depth, max-depth, ...)
- Optional filtering mechanisms in order to remove spurious data that may introduce distortion in the offset computations (swath & depth filters, stand-alone version of a despiking algorithm [5], ...)
- Integration in the XYZ computation for each track, of the recorded sound velocity profiles and tide measurements.
- XTF and QPS-DB files import, ASCII export of both the XYZ and raw data (position, attitude, measured slant ranges)



(a) Without calibration



(b) After calibration

Fig. 10. Extracting profiles

FURTHER DEVELOPMENTS & APPLICATIONS

While the patch test program core consists of a fully automated algorithm able to compute the full set of offsets, its interface offers the possibility for the user to finely control it. Thus, the advantages of both approaches (graphical & automatic) have been combined and associated to powerful functionalities, in order to deliver the full acquisition context to the operator. In very different environments, this tool has been successively tested on various bathymetric sensors.

Using the patch test program may improve the reliability of various applications (DTM computations, data fusion with other sensors such as side-scan) by supplying them with more accurate raw information [6].

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